

Methyl Ethyl Ketone Production



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Senior Design, Mini Project 4

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Project Charter

Title	Goal	In Scope	Out of Scope
Methyl Ethyl Ketone Production	The goal of this project is to produce 10,000 metric tons/yr at plant operating rate of 8,000/yr	• Capital cost with order of magnitude • Equipment costs (ASPEN & Hand calculation) • ISBL, OSBL, contingency, engineering costs, total capital costs • Variable cost production (raw materials and products) • Fixed cost production, revenues, and gross profit, payback period and ROI • NPV for 7- and 5-year recovery term • Discounted cashflow return using MACRS depreciation w/ 5- and 7-year recovery term • Sensitivity analysis on utilities, raw material cost and product cost	• ASPEN Simulation • Heat Integration • Pinch Analysis • Piping, sizing or the process
Deliverables		Team Members	
1. Total capital cost 2. Total cost of equipment 3. Variable cost production 4. Fixed cost production 5. NPV for 7- and 5-year recovery term 6. Discounted cashflow return using MACRS depreciation w/ 5- and 7-year recovery term 7. Sensitivity analysis on utilities, raw material cost and product cost		• Huda Usman • Nisha Patel • Alexis Miranda • Courteney Fabbri	

Design Plan

Economic Analysis of Methyl Ethyl Ketone Production					
Task ID	Task	Start Date	Duration (Days)	Finish Date	Assignee
Logistics		Nov 12	5	Nov 17	
1	First Meeting (Task Delegation)	Nov 12	1	Nov 13	All
2	Project Charter	Nov 13	1	Nov 14	Huda
3	Design Plan	Nov 13	1	Nov 14	All
16	Layout/Format	Nov 16	1	Nov 17	All
Execution		Nov 13	3	Nov 16	
Design Basis					
4	Objective and project description	Nov 13	1	Nov 14	Alexis
Economic Analysis of MEK Process					
5	Order of magnitude method	Nov 13	1	Nov 14	Alexis
6	Equipment cost	Nov 13	1	Nov 14	Alexis, Courteney
7	ISBL, OSLB, Contingency, Engineering Cost	Nov 14	1	Nov 15	Alexis, Nisha
8	Total capital cost	Nov 14	1	Nov 15	Alexis, Nisha
9	Variable cost of production	Nov 14	1	Nov 15	Courteney, Nisha
10	Fixed cost of production/Total Capital Costs	Nov 14	1	Nov 15	Nisha, Huda
11	Revenues and gross profit	Nov 15	1	Nov 16	Huda, Courteney
12	Simple pay-back, & ROI	Nov 15	1	Nov 16	Alexis, Huda, Courteney
13	NPV value for 5- & 7- year term	Nov 15	1	Nov 16	Huda
14	DCFROR for 5- & 7- year term	Nov 15	1	Nov 16	Huda
15	Sensitivity Analysis	Nov 15	1	Nov 16	Nisha, Huda
Wrap-up					
17	Section Unification	Nov 16	1	Nov 17	All
18	Proofreading / Plagiarism Check	Nov 16	1	Nov 17	All

Process Description

The objective of this report is to analyze the economics of a Methyl ethyl ketone (MEK) production process.

The process will use 2-butanol to produce 10,000 metric tons of MEK per year. The feed stock is pumped through a heater into a vaporizer. From here, the vapors are heated and fed into a reaction system with a 90% conversion rate to MEK, hydrogen, and other products. MEK is then separated using absorption, extraction, recovery, and purification columns.[11]

The plant is expected to operate 8,000 hours per year and will be built upon an existing site that already has adequate infrastructure to provide for any extra requirements the plant may need.

Part A) Capital Cost With Order of Magnitude Method:

The order of magnitude estimates are class 5 estimates that are made at the initial stages of the project. These methods consider the past data for a similar process and adds a scaling factor based on the capacities of the plants to make a good estimate for the cost of the process [4]. Due to limited information regarding the cost data of a similar Methyl Ethyl Ketone project we will be analyzing the capital cost using Bridgewater's method. Bridgewater's method is an order of magnitude estimate that considers the number of processing steps and functional units.[9]

Since our MEK plant has a production goal of 10,000 metric tons per year. Since our production rate is well below the Bridgewater methods threshold of 60,000 metric tons per day, we will be using the equation below:

$$C = 380,000N \left(\frac{Q}{s} \right)^{0.3}$$

The ISBL capital cost, C, is found by using the number of functional units, N, the plant capacity, Q, and the reactor conversion, s. The proposed MEK plant has a total of 17 functional units and the reactor is expected to have a 90% conversion. Therefore, the ISBL capital cost is \$105.7 MM $\pm$ 50% by Bridgewater's method for order of magnitude estimates.

Part B) Equipment Costs Using ASPEN & Hand Calculation:

This section will compare calculating equipment costs by hand and by using ASPEN's cost estimator software as well as explain the assumptions made for the equipment being examined. The total cost has been adjusted for inflation for both methods using Marshall & Swift cost indices. The index for year 2020 is based on a 40% increase from 2010.[6]

Marshall Swift index 2020 = 2025.2

Marshall Swift index 2013 = 1552.8

Marshall Swift index 2010 = 1446.5

Table 1 below shows the hand-calculated total equipment cost for each unit used in the production process. The total cost includes equipment cost and installation as well as adjustments made for using stainless steel as the equipment material.

Several assumptions were made when calculating the cost for certain equipment. Correlation data was used based on the cost curves for purchased equipment costs. This correlation data was found in Towler's "Chemical Engineering Design" and used in the equation below to estimate the equipment cost. Installation for equipment was calculated using Hand's installation factors. The cost of equipment was converted from carbon steel to stainless steel using a materials factor of 1.3. [9]

$$C_e = a + bS^n$$

Assumptions:

- For all pressurized vessels and columns, a plate thickness was assumed to be 0.75in.
- For all stainless-steel units, it was assumed that stainless steel 304 was the grade used for the process.
- Cost per tray for separation columns is based on a stack of 30 trays.
- The density of carbon steel and stainless steel 304 is 7900 kg/m³ and 8000 kg/m³ respectively.[9]
- Feed compressor cost includes cost for an explosion proof motor.
- Shell mass for columns and pressurized vessels was found using the vessels diameter, length, wall thickness, and the density of the material being used.[9]

$$Shell\ mass\ (kg) = \pi D_c L_c t_w \rho$$

Table 1: Hand calculated equipment costs		
ID	Equipment Name	Total Cost and adjustments, \$ (2010)
1	Butanol Vaporizer	117,518.07
2	Reactor Feed Heaters (Qty: 2)	278,186.39
3	Reactors (Qty: 3)	462,791.72
4	Condenser	139,093.19
5	Absorption Column	111,720.67
6	Extraction Column	112,432.43
7	Solvent Recovery Column	122,631.63
8	Recovery Column Reboiler	108,361.98
9	Recovery Column Condenser	19,775.00
10	Solvent Cooler	31,395.00
11	Product Purification Column	835,368.76
12	Product Column Reboiler	138,287.61
13	Product Column Condenser	153,811.43
14	Feed Compressor	4,380,634.82
15	Butanol Storage Tank	279,656.32
16	Solvent Storage Tank	169,831.25
17	Product Storage Tank	279,656.32
Total Cost, \$ (2010, ±30%):		7,741,152.60
Total Cost, \$ (2019, ±30%):		10,838,148

ASPEN equipment cost calculations were found for the butanol vaporizer, recovery column condenser, and the product purification column. The results are presented in tables 2, 3, and 4 below.

Table 2: ASPEN Butanol Vaporizer Summary Costs \$(2013)

Item	Material (USD)	Manpower (USD)	Manhours
Equipment & Setting	15000	953	32
Piping	4984	11831	389
Civil	995	1490	62
Structural Steel	0	0	0
Instrumentation	13172	6207	203
Electrical	0	0	0
Insulation	2082	2754	121
Paint	356	1042	46
Subtotal, \$(2013)	36589	24277	853
Subtotal, \$(2020)	47719	31662	1112

Total material and manpower cost, \$(2013) = USD 60900

Total material and manpower cost, \$(2020) = USD 79426

For the equipment cost of the butanol vaporizer unit, the ASPEN calculations differed from the hand calculations by 51.7%. Inputs used in the ASPEN Capital Cost Estimator can be referenced in Table 1 in the appendix. This could be caused by many factors due to the class 4 nature of the cost estimate. One reason may be the year of the data that was used, ASPEN was calculated in 2013 while the hand calculations used 2010 data. Another reason may be due to assumptions made during hand calculations. The vaporizers design was assumed to be a carbon steel u-tube kettle reboiler with a heat transfer area of 15 m², other parameters were not considered.

Table 3: ASPEN Recovery Column Condenser Summary Costs \$(2013)

Item	Material (USD)	Manpower (USD)	Manhours
Equipment & Setting	2700	472	16
Piping	15314	11040	367
Civil	942	1426	59
Structural Steel	0	0	0
Instrumentation	8007	2152	69
Electrical	0	0	0
Insulation	3571	3214	141
Paint	383	993	44
Subtotal, \$(2013)	30917	19297	696
Subtotal, \$(2020)	40321	25167	908

Total material and manpower cost, \$(2013) = USD 50200

Total material and manpower cost, \$(2020) = USD 65471

The equipment cost for the recovery column condenser on ASPEN differed from the hand calculation by more than 100%. Inputs used in the ASPEN Capital Cost Estimator can be referenced in Table 2 in the appendix. This large margin of error is partly due to the difference in the year the data was used to calculate each cost. However, most of the difference arises from the assumptions made when calculating by hand. This heat exchanger was assumed to be a carbon steel double pipe heat exchanger with its only parameter of consideration being the heat exchange area. ASPEN most likely estimates the cost using by assuming a greater number of parameters and thus resulted in a more expensive estimate.

Table 4: ASPEN Product Purification Column Summary Costs \$(2013)

Item	Material (USD)	Manpower (USD)	Manhours
Equipment & Setting	151100	1670	54
Piping	49127	21499	718
Civil	2485	2983	121
Structural Steel	12742	2401	85
Instrumentation	33440	13031	424
Electrical	2746	1509	51
Insulation	14133	12218	535
Paint	349	622	28
Subtotal, \$(2013)	266122	55933	2016
Subtotal, \$(2020)	347076	72948	2629

Total material and manpower cost, \$(2013) = USD 322100

Total material and manpower cost, \$(2020) = USD 420083

Separation columns have several components associated with them such as type of packing, number of trays, and the type of vessel it is. The purification column is a stainless-steel plate column with stainless steel sieve plates. For hand calculations the cost per tray was based on a stack of 30 trays when the column itself only required 15 trays. The cost of the column was estimated based on its shell mass using stainless steel 304. An installation factor of 4 was used as well, increasing the cost to be 64.1% greater than the ASPEN calculation. Inputs used in the ASPEN Capital Cost Estimator can be referenced in Table 3 in the appendix.

Part C) ISBL, OSBL, Contingency, Engineering Costs:

The total installed ISBL cost of the plant was found to be \$10.8 MM $\pm$ 30% using cost curve correlations.

Since the plant will be built on an existing site that satisfies all requirements for the new plant there is no need for extensive offsite considerations. Therefore, the OSBL was taken to be only 20% of the total installed ISBL cost.

Contingency is the extra costs that allows for variation in the estimated costs and accounts for labor disputes, weather related problems, currency fluctuations, change in prices, etc. For this plant, it was assumed to have the minimum contingency and so it was taken to be 10% ISBL plus OSBL.

The engineering cost was determined to be best estimated at 30% of the total installed ISBL cost due to the uncertainty in pricing of contractor and construction services.

Working capital was estimated to be 15% of the total installed ISBL cost since it is the cost of beginning plant operation.

Table 5: Capital cost breakdown

Capital Component	Reference Calculation	Cost, \$(2020)
Total ISBL		10,838,148
OSBL	20 % of ISBL	2,167,630
Contingency	10 % of ISBL + OSBL	3,251,445
Engineering Cost	30 % of ISBL + OSBL	5,419,074
Working Capital	15% of ISBL + OSBL	3,793,352
Total Capital Cost		25,469,650

Part D) Total Capital Cost:

The total capital cost is the sum of ISBL, OSBL, working capital, engineering cost, and contingency cost. All together the estimated cost of the MEK manufacturing plant is \$20,158,956 as shown in Table 5.

Part E) Variable Cost of Production (Raw Materials and Utilities):

The variable cost of production is the sum of all component costs in production minus the by-product revenue. Because the by-product, hydrogen, was assumed to be 100% recycled back into our feedstock, by-product revenue was zero. Therefore, the total VCOP was the sum of raw material cost and utilities.

To calculate the cost of raw materials, pricing data needed to be collected for the feedstock and solvent. The feedstock 2-butanol was found to be \$800/tonne in 2012 according to Towler [9]. The solvent chloroform was found to be \$639/tonne in 2009 according to ICIS [8]. Given the rates of consumption from the problem statement, cost of both raw materials were calculated and converted to \$(2020), as presented in Table 6. Total cost of raw materials was \$11,024,533.

Table 6: Raw Materials Costs

Chemical	Usage	Avg price/unit	Pricing year	Cost in pricing year	Cost, \$(2020)
2-butanol	10.45 KMTA	\$800/tonne ^[9]	\$(2012)	8,360,000	11,018,442
Chloroform	7000 kg/year	\$639/tonne ^[8]	\$(2009)	4473	6091
Total cost, \$(2020):			\$11,024,533		

Next, utilities for production were calculated. The following unit conversions for a metric tonne were used in calculating utility cost: 1 metric tonne = 264.17 US gallon of water = 2204.64 lbs. Furthermore, it was assumed in calculations that 1 GJ = 947,817.12 Btu [13]. Next, assuming an operating year consisted of 8,000 hours, Table 7 was tabulated using average pricing of units in

Figure 1. The total cost of utilities was converted to \$(2020) using a Marshall and Swift index of 1.4 [6]. The total utility cost was calculated to be \$2,595,368.

1. Electricity:	0.09 USD/kWh
2. HP Steam:	11.45 USD/1000lb
3. MP Steam:	9.50 USD/1000lb
4. LP Steam:	8.48 USD/1000lb
5. Cooling Water:	0.08 USD/1000gal
6. Fuel Gas (LHV)	7.30 USD/MMBtu

Figure 1. Standard utility prices for 2010

Table 7: Utilities cost				
Utility	Usage	Avg price/unit	Cost, \$(2010)	Cost, \$(2020)
Fuel oil	135,000 GJ/year	\$7.30/MMBtu ^[13]	934,071	1,307,699
Cooling water	960 KMTA	\$0.08/1000gal ^[13]	20,288	28,404
Steam, low pressure	9.6 KMTA	\$8.48/1000lb ^[13]	179,475	251,265
Power	1 MW/year	\$0.09/kWh ^[13]	720,000	1,008,000
Total cost (\$2020)		\$2,595,368		

Using a total raw material cost of \$11,024,533 and total utility cost of \$2,595,368, the variable cost of production VCOP was calculated to be **\$13,619,901**.

Forecasted prices for 2-butanol, chloroform, and MEK are targeted by Venture Capital News Network (VCNN) to take different trends. According to VCNN, n-butanol based oxo alcohol prices are projected to increase up to 5% by the year 2024. Currently, the price of 2-butanol is estimated to be \$1054/tonne, so the rise will increase prices to \$1107/tonne. This is attributed to the demand in the market for perfumes, paints, coatings, dyes, synthetic resins, and alkaloids in upcoming years [16]. Furthermore, competition for renewable biofuel such as bio-diesel used in combustion engines is also forecasted to increase demand of 2-butanol in the future.

Furthermore, according to GlobeNewswire, the United States chloroform market is expected to maintain a 2.5% growth rate by the year 2025 [17]. Currently, the price of chloroform is estimated to be \$870/tonne, so this rise will increase prices \$892/tonne. GloveNewswire states this momentum is a steady rise previously seen in the US market, so growth isn't attributed to any distinguishing factors.

Finally, VCNN forecasts MEK prices to rise 4.5% over the period from 2018-2024 [18]. Currently, the price of MEK is estimated to be \$2.63/kg, so this rise will increase prices \$2.75/kg. Similarly to oxo alcohols, this projected increase in prices is due to the rise in construction expenditure. Consumer demand for products such as paints and coatings applications as well as anesthetics, antiseptics, and lotions is expected to increase prices in the next 5 years.

Part F) Fixed Cost of Production (Assume That A Year Equals 8000 Hours. This Is About 330 Days, Which Allows For Periodic Shut-Down And Maintenance):

The salary provided by the companies differ by different regions. Here we assumed that our plant is based in Gulf Coast USA where the average salary was found to be \$50,000/annum [1]. Also, we assumed that there are 5 operators at each position of the shift and total shifts were taken to be a total of 3. The following equation is then utilized to calculate the operational labor taking the above information into the account and it was found to be **\$750,000**.

$$\text{Operational labor} = (\text{Operators per shift}) * (\text{Number of shift positions}) * \text{Salary per year}$$

Taking an account for the periodic shutdown and maintenance, components of the fixed cost of production for Gulf Coast USA were calculated. The supervision was taken to be 25% of the operating labor and the direct salary overhead was taken to be 50% of the operating labor plus the supervision. The results are shown in the table below.

Table 8: Fixed Cost of Production Calculations

Operating Labor		\$750,000
Supervision	25% of Operating Labor	\$187,500
Direct Salary Overhead	50% of Operating Labor plus Supervision	\$562,500

While calculating the maintenance, land rent, property tax and insurance, and the allocated environmental charges, ISBL and OSBL cost percentages are taken into consideration. For this project, we have taken maintenance to be 3% of the ISBL, rent and allocated environmental charges are taken 1% of the ISBL plus OSBL, and property taxes and insurance is taken to be 1% of ISBL. Total fixed cost of production was also calculated by taking the above factors in account. The summary of the results is shown in the table below.

Table 9: Total Fixed Capital Cost

COSTS COMPONENTS		VALUES
Total ISBL		\$10,838,149
Maintenance (Material + Labor)	3% of ISBL	\$325,144
Rent of land/buildings	(1% of ISBL) + OSBL	\$2,276,011
Property taxes and insurance	1% of ISBL	\$3,251
Allocated environmental charges	(1% of ISBL) + OSBL	\$2,276,011
Average salary (per year) of an operator		\$50,000
Shift positions		3
Number of operators per shift position		5
Operating Labor		\$750,000
Supervision	25% of Operating Labor	\$187,500
Direct Salary Overhead	50% of Operating Labor plus Supervision	\$562,500
Total Fixed Capital Cost		\$6,380,418

Therefore, the total fixed capital cost was found to be **\$ 6.4 MM**.

Part G) Revenues and Gross Profit:

To determine the process profitability, it is essential to evaluate the product revenues and costs of production. The revenues are the income that comes from selling the process main products and by-products [9]. In this project, the main product is methyl ethyl ketone (MEK) and by-product is hydrogen. We are taking a design basis for production rate of MEK to be 10 MM kg per year.

Gross Profit:

The gross profit takes an account for all other variable costs, including raw materials, fixed costs, and by-product revenues. The gross profit is given as,

$$\text{Gross Profit} = \text{Main product revenues} - \text{CCOP}$$

Where CCOP is the cash cost of production and can be calculated as,

$$\text{CCOP} = \text{VCOP} + \text{FCOP}$$

Where, VCOP is the sum of all variable costs of production minus by-product revenues

FCOP is the sum of the fixed costs of production

The sale price of MEK is given to be \$2/kg, and the target is to produce 10 KMTA; therefore, selling the product will earn \$20 MM per year, \$(2012). Using the Marshall & Swift index for \$(2020) over \$(2012), the total product revenue in \$(2020) was calculated to be \$26.36 MM. Next, the gross profit was calculated by subtracting fixed capital cost FCC from the main product revenue. The results are summarized in the Table X shown below, where total gross revenue was calculated to be \$6,359,590.

Table 10: Results for Gross Profit Calculation

Gross Profit = Main product revenues – CCOP		
CCOP = VCOP + FCOP		
VCOP (\$2020)	FCOP (\$2020)	CCOP (\$2020)
\$13,619,901	\$6,380,418	\$20,000,319
Gross Profit (\$2020)		
\$6,359,589.68		
Main Product Revenue (\$2020)	\$26,359,909	

Part H) Simple Payback Period and ROI:

Simple payback time and ROI are the part of simple methods for economic analyzer, it quickly estimates the economic measures with known cash flows and investments and estimates the time when the plant will start operating at full capacity i.e. 100 %. For simple payback period, taxes and depreciation is neglected and it is based on average annual cash flow and the total investment. Simple payback time is given as,

$$\text{Simple payback time} = \frac{\text{total investment}}{\text{average annual cashflow}}$$

The total investment will be the capital cost which is found to be \$ 25.47 MM for MEK production. Also, the average annual cashflow can be calculated by subtracting total cash inflows (income by selling products and by-products) and outflows (CCOP).

$$\text{Simple payback time} = \frac{\$ 25.47 \text{ MM}}{\$6.36 \text{ MM}} = 4.00$$

To calculate the total time, we must include plus two and a half years in our calculations to take an account for the years after which the plant will start making money. Therefore,

$$\text{Simple payback time} = 4 + 2.5 = 6.5 \text{ years}$$

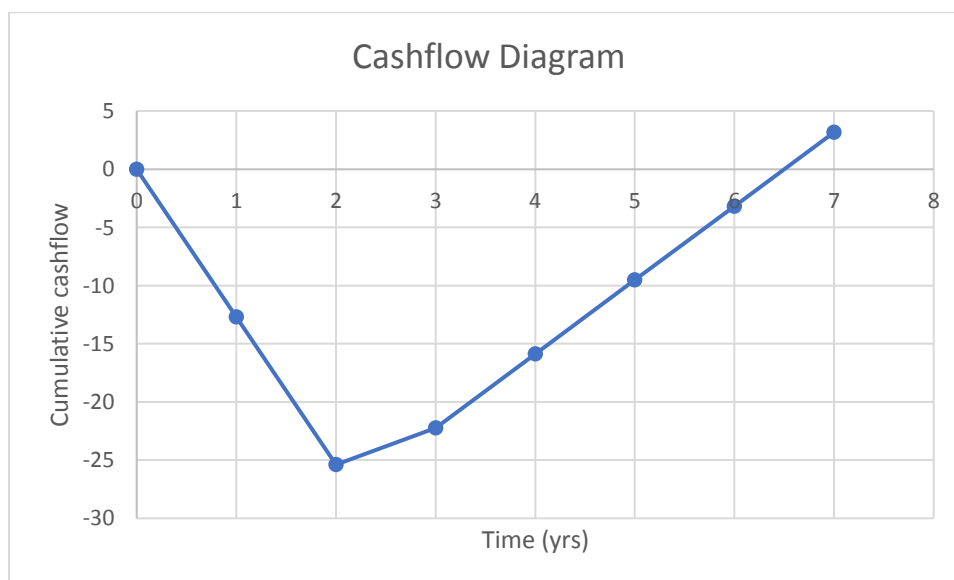


Figure 2. Project cashflow diagram

When finding the simple payback time graphically it was found to be 6.5 years. This evaluation method includes period of construction and the 50% production rate during the first year of operation.

Return on Investment (ROI)

The return on investment is a measure of economic performance that depends on the net annual profit, and the investments made toward the plant. ROI indicates the benefit of an investment, and how effective the plant will generate a profit from its investments.

$$ROI = \frac{\text{Net Annual Profit}}{\text{Total Investment}}$$

Net annual profit is the gross profit subtracted by taxes. According to IRS Publication 542, tax years beginning after the year 2017 for corporations will determine tax by multiplying taxable income by 21% [14]. The expression below calculates the net annual profit used for the ROI calculation.

$$\text{Net Annual Profit} = \text{Gross Profit} \left(1 - \frac{\text{Tax Rate}}{100} \right)$$

$$\text{Net Annual Profit} = \$6.36 \text{ MM} (1 - 0.21) = \$5.02 \text{ MM}$$

After obtaining the net annual profit of \$5.02 MM, it is possible to calculate the ROI post-tax. This expression is presented below as the net annual profit of \$5.02 MM divided by the total investment of \$25.47 MM.

$$ROI = \frac{\text{Net Annual Profit}}{\text{Total Investment}}$$

$$ROI = \frac{\$5.02 \text{ MM}}{\$25.47 \text{ MM}} = 0.1971$$

The pre-tax ROI follows cashflow only and can be used to eliminate the effect that depreciation has on annual profit. This value is obtained by taking pretax revenue—gross profit—and dividing it by the total investment. The value of ROI obtained pre-tax was 0.2497 as presented below.

$$\text{Pretax ROI} = \frac{\text{Pretax revenue}}{\text{Total Investment}}$$

$$\text{Pretax ROI} = \frac{\$6.36 \text{ MM}}{\$25.47 \text{ MM}} = 0.2497$$

According to Trendshare, “a really good return on investment for an active investor is 15% annually” [15]. With such a high ROI, the plant will do well in utilizing its investments and produce more gains in profit than losses.

Part I) NPV Value Calculate at 14 % Discount Rate Using MACRS Depreciation With 7 and 5 Year Recovery Term:

Net Present Value (NPV) is defined as the difference between the present value of cash inflows and the outflows over time. It was also be defined as the sum of the present values of the future cash flows. If the NPV is positive, it is assumed that the made investment will be profitable. On the other hand, if the NPV is negative, an investment will result in a net loss [10].

Modified Accelerated Cost Recovery System (MACRS) was introduced in the U.S to measure the depreciations in the investments for any processing plant. For this project, we are assuming that the fixed capital cost will be paid in equal increments for first two years. For the third year, there is 50 % production i.e. the variable cost of production and product revenues will be reduced to half of their initial value. Therefore, the pre-tax revenue for the third year can be calculated by taking the difference of ½ of the variable cost and the total fixed capital cost. After the first three years, we are assuming that the plant is operating at the production rate of 100 %. From the above said information, pre-tax cash flow values are input in the table. Depreciation values can be found on the IRS website for 5 and 7 year schedules. The depreciation value is then multiplied with the capital cost of the plant and divided by a hundred. Taxable income can then be calculated by taking the difference between pre-tax cash flow and the depreciation. All non-positive values for taxable income are taken as zero. The tax was assumed to be imposed at 35 %. As it can be seen in the tax due column, at year 5th the value for the taxable income is positive however the tax due is zero. This is because, we have to pay the tax in the following year for the

income we earned the same year. Next, the cash flow can be calculated by subtracting tax due from the pre-tax cash flow. It can be observed that initially plant's revenue was in negative which makes sense because the plant is not making money but eventually, the values are starting to go up. The discounting factor "Fd" can be calculated by $1/(1+i)^n$ where "i" is the interest rate and "n" represent number of years. Discounted cash flow helps determine the present value of the expected future cash flows using a discounted rate [10]. The present value is then calculated by multiplying the cash flow and the discounted cash flow. In this project, we are assuming plant life to be 15 years therefore, the NPV is then calculated by simply adding the present values up to 15 years.

Table 11: NPV with 5 years MACRS - 14 % discount rate

year	Pre-tax CF	depreciation	taxable income	tax due	Cash flow	Fd	Present Value	NPV
1	-12.735	0	-12.735	0	-12.735	0.877193	-11.1711	-11.1711
2	-12.735	0	-12.735	0	-12.735	0.769468	-9.79917	-20.9702
3	3.62	4.226	-0.606	0	3.62	0.674972	2.443397	-18.5268
4	6.36	6.7616	-0.4016	-0.2121	6.5721	0.59208	3.891211	-14.6356
5	6.36	4.05696	2.30304	-0.14056	6.50056	0.519369	3.376187	-11.2594
6	6.36	2.434176	3.925824	0.806064	5.553936	0.455587	2.530299	-8.72913
7	6.36	2.434176	3.925824	1.374038	4.985962	0.399637	1.992576	-6.73655
8	6.36	1.217088	5.142912	1.374038	4.985962	0.350559	1.747874	-4.98868
9	6.36	0	6.36	1.800019	4.559981	0.307508	1.40223	-3.58645
10	6.36	0	6.36	2.226	4.134	0.269744	1.115121	-2.47133
11	6.36	0	6.36	2.226	4.134	0.236617	0.978176	-1.49315
12	6.36	0	6.36	2.226	4.134	0.207559	0.858049	-0.6351
13	6.36	0	6.36	2.226	4.134	0.182069	0.752675	0.117574
14	6.36	0	6.36	2.226	4.134	0.15971	0.660241	0.777815
15	6.36	0	6.36	2.226	4.134	0.140096	0.579159	1.356974

Similarly, same procedure is applied for depreciation of the 7 years MACRS to calculate pre-tax cash flow, taxable income, tax due, cash flow, discounting factor, present value and finally, net present value. Upon comparison, we found that the NPV value for 7 years was lower as compared to 5 years. Also, for 7 years, we had depreciation values and they were smaller as compared to 5 year's depreciation values. It can be concluded that higher the NPV value, higher will be the profit of the plant.

Table 12: NPV with 7 years MACRS - 14 % discount rate

year	Pre-tax CF	depreciation	taxable income	tax due	Cash flow	Fd	Present Value	NPV
1	-12.735	0	-12.735	0	-12.735	0.87719	-11.17105	-11.17105
2	-12.735	0	-12.735	0	-12.735	0.76947	-9.799169	-20.97022
3	3.62	3.019477	0.600523	0	3.62	0.67497	2.4433969	-18.52682
4	6.36	5.174737	1.185263	0.21018305	6.14981695	0.59208	3.6411853	-14.88563
5	6.36	3.695637	2.664363	0.41484205	5.94515795	0.51937	3.0877287	-11.79791
6	6.36	2.639137	3.720863	0.93252705	5.42747295	0.45559	2.4726837	-9.325226
7	6.36	1.886909	4.473091	1.30230205	5.05769795	0.39964	2.0212449	-7.303982
8	6.36	1.886909	4.473091	1.56558185	4.79441815	0.35056	1.6807267	-5.623255
9	6.36	1.886909	4.473091	1.56558185	4.79441815	0.30751	1.4743217	-4.14893
10	6.36	0.942398	5.417602	1.56558185	4.79441815	0.26974	1.2932646	-2.855669
11	6.36	0	6.36	1.8961607	4.4638393	0.23662	1.0562219	-1.799447
12	6.36	0	6.36	2.226	4.134	0.20756	0.8580493	-0.941397
13	6.36	0	6.36	2.226	4.134	0.18207	0.7526749	-0.188723
14	6.36	0	6.36	2.226	4.134	0.15971	0.6602411	0.471518
15	6.36	0	6.36	2.226	4.134	0.1401	0.5791589	1.0506769

Part J) Discounted Cash Flow Rate of Return Using MACRS Depreciation with 7 and 5 Year Recovery Term:

The discounted cash flow rate of return DCFROR was calculated using the “goal seek” in the excel sheet. The NPV value was targeted to be equal to zero by changing the selected cell of the discount rate. After running the goal seeking method, the interest rate for 5 year was found to be **15.32 %** and for 7 year it was found to be **15.00 %**. Since the recovery term is higher for the 5-year project plant, it was concluded that the **most profitable project plant for MACRS depreciation will be with 5-year recovery term**. The results of analysis can be seen in Appendix I.

Part K) Sensitivity Analysis on Utilities, Raw Materials Cost and Product Cost:

The economic analysis can be studied on base of the calculated estimates of investments and cash inflow and out flow of a plant [9]. Sensitivity analysis of any plant is a way of studying what could be the effects of some uncertainties on viability of a project. The graphs obtained below shows that the NPV is directly dependent on the cost of raw materials i.e. variable cost of production whereas, its not so dependent on the cost of utilities.

Another factor that can affect NPV is the cost of the main product i.e. MEK. The price of MEK affects the main product revenues which then directly changes the Gross profit.

Therefore, it can be said that the profits of the plant are highly dependent on prices of raw materials at which they are bought and the price of the product i.e. MEK at which it will be sold. Although its not much dependent on the operational cost which covers the utilities cost.

The graphs show that if the raw material prices are increased any more than 40%, then the NPV values for both 5- and 7-years terms will go negative. Additionally, if the product selling price goes any below than 60% then the NPV will be negative for both the years term. Hence, it can be concluded that the optimum operational limit for the estimated MEK plant is no less than 60% decrease in the product price, and no more than 40% increase in the buying price of the raw materials.

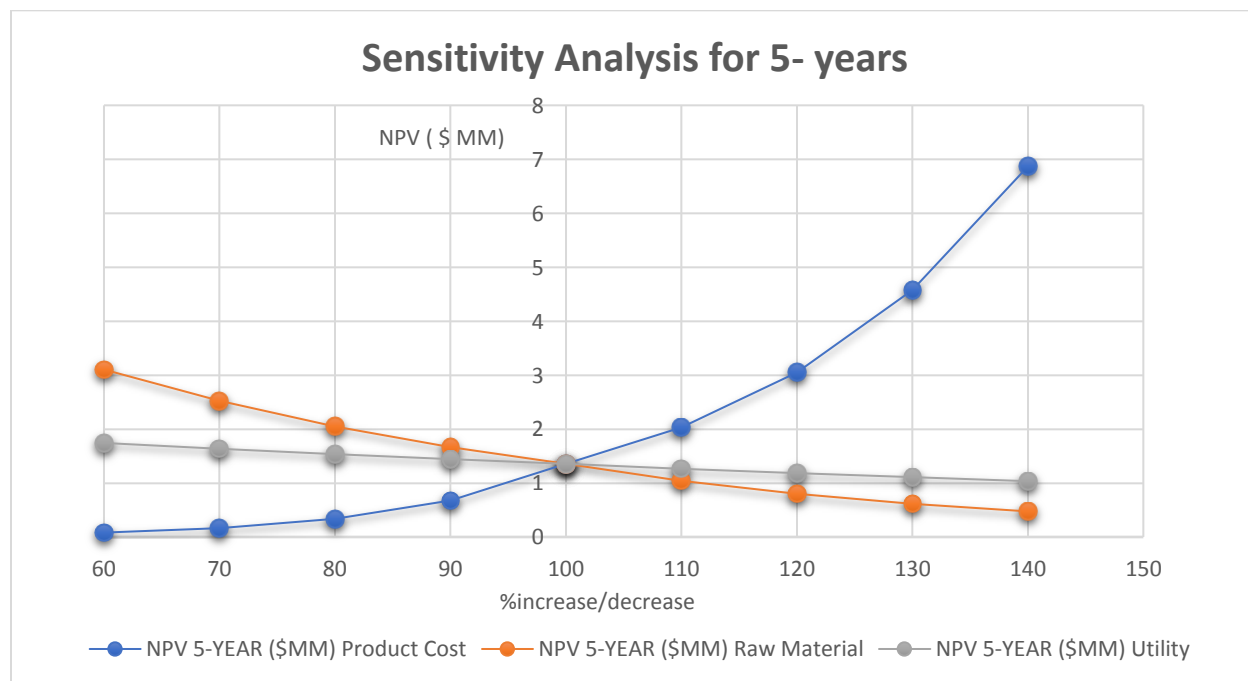


Figure 3. Sensitivity Analysis for 5 recovery years

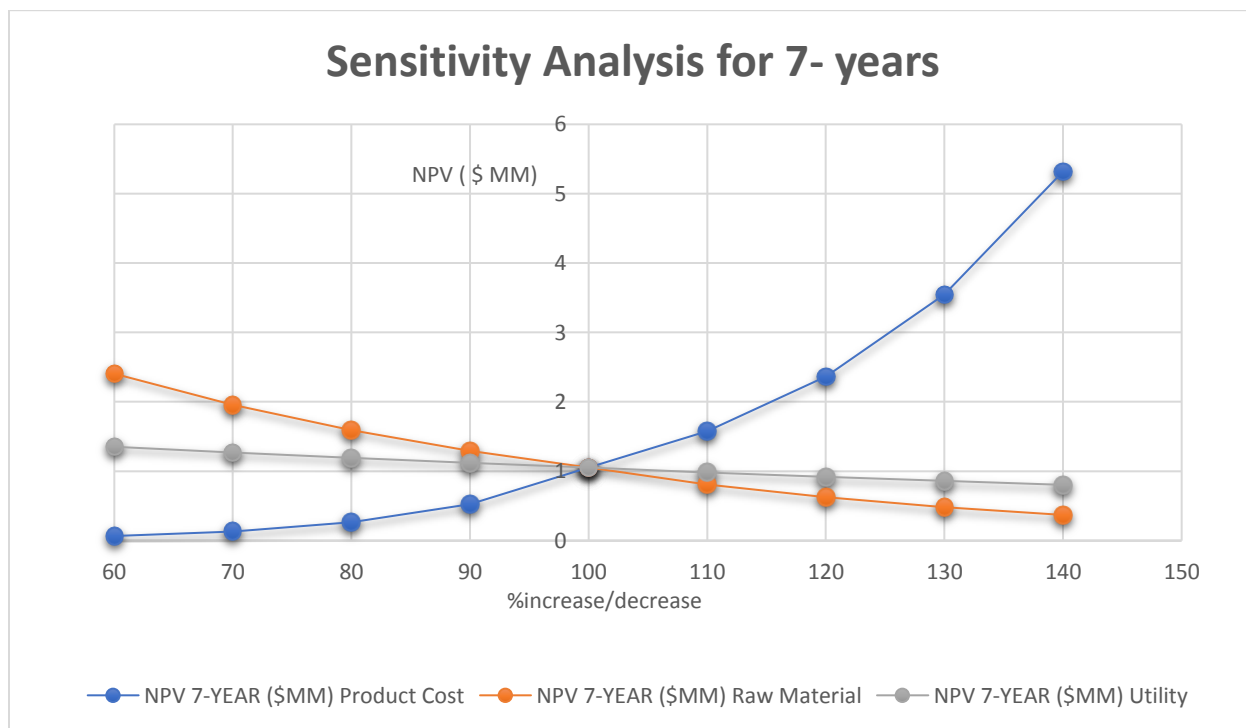


Figure 4. Sensitivity Analysis for 7 recovery years

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Appendices: Tables and Graphs

Table 1. ASPEN Capital Cost Estimator Output for Butanol Vaporizer
BUTANOL VAPORIZER

Item Code: DRB KETTLE
Internal Name : DRB KETTLE BUTANOL VAPORIZER
[Sizing Data](#)
[Design Data](#)
[Summary Costs](#)

Sizing Data		
Description	Value	Units
Design Data		
Parameter	Value	Units
Item type	KETTLE	
Number of identical items	1	
GENERAL DESIGN DATA		
TEMA type	BKT	
Heat exchanger design option	STAND	
Heat exchanger design+cost tool	ECON	
Heat transfer area	15.000	M2
Number of shells	1	
Number of tube passes	2	
Number of shell passes	1	
Vendor grade	HIGH	
SHELL DATA		
Shell material	A285C	
Shell diameter	425.000	MM
Shell length	4.0000	M
Shell design gauge pressure	500.000	KPAG
Shell design temperature	340.000	DEG C
Shell operating temperature	100.000	DEG C
Shell corrosion allowance	3.0002	MM
Shell wall thickness	9.5250	MM
ASA rating Shell side	150	CLASS
Number of baffles	9	
Shell fabrication type	PIPE	
Expansion joint	NO	
TUBE DATA		
Tube material	A 214	
Number of tubes per shell	63	
Tube outside diameter	25.000	MM
Tube port diameter	300.000	MM
Tube length extended	3.0000	M
Tube design gauge pressure	500.000	KPAG
Tube design temperature	340.000	DEG C
Tube operating temperature	100.000	DEG C
Tube corrosion allowance	0.060002	MM
Tube wall thickness	1.7000	MM
Tube gauge	16	BWG
Tube pitch symbol	TRIANGULAR	
Tube pitch	32.000	MM
Tube seal type	SEALW	
TUBE SHEET DATA		
Tube sheet material	A 515	
Tube sheet thickness	20.000	MM
Tube sheet corrosion allowance	3.0001	MM
Channel material	A 515	
PROCESS DESIGN DATA		
Duty	0.189275	MEGAW
Heat of vaporization	647.580	KJ/KG
Vaporization	90.000	PERCENT
Specific gravity tower bottoms	0.500000	
Molecular weight Bottoms	74.122	
HEAD DATA		
Head material Tube side	A 515	
ASA rating Tube side	150	CLASS
Head thickness Tube side	7.0002	MM
WEIGHT DATA		
Shell	410	KG
Tubes	190	KG
Heads	37	KG
Internals and baffles	18	KG
Nozzles	60	KG
Flanges	60	KG

Base ring and lugs	6	KG
Tube sheet	22	KG
Saddles	35	KG
Total weight	850	KG
VENDOR COST DATA		
Material cost	2565	DOLLARS
Shop labor cost	3646	DOLLARS
Shop overhead cost	3789	DOLLARS
Office overhead cost	2264	DOLLARS
Profit	2736	DOLLARS
Total cost	15000	DOLLARS
Cost per unit weight	17.647	USD/KG
Cost per unit area	1000.000	USD/M2

Summary Costs

Item	Material(USD)	Manpower(USD)	Manhours
Equipment&Setting	15000.	953.	32
Piping	4984.	11831.	389
Civil	995.	1490.	62
Structural Steel	0.	0.	0
Instrumentation	13172.	6207.	203
Electrical	0.	0.	0
Insulation	2082.	2754.	121
Paint	356.	1042.	46
Subtotal	36589	24277	853

Total material and manpower cost=USD 60900.

Table 2: ASPEN Capital Cost Estimator Output for Product Purification Column;

PROD PURIFICATION COLUMN

Item Code: DTW TRAYED

Internal Name : DTW TRAYED PROD PURIFICATION COLUMN

[Sizing Data](#)

[Design Data](#)

[Summary Costs](#)

Sizing Data

Description	Value	Units
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Design Data

Parameter	Value	Units
Item type	TRAYED	
Number of identical items	1	
EQUIPMENT DESIGN DATA		
Tray type	SIEVE	
Application	DIS-RB	
ASME design basis	D1NF	
Design gauge pressure	200.000	KPAG
Design temperature	340.000	DEG C
Operating temperature	305.000	DEG C
GENERAL DESIGN DATA		
Number of platforms	3	
COLUMN DATA		
Shell material	SS316	
Diameter option	OD	
Vessel diameter	1.0000	M
Vessel tangent to tangent height	20.000	M
Head type	TORI	

MECHANICAL DESIGN DATA		
Wind or seismic design	W+S	
Fluid volume	20.000	PERCENT
Weld efficiency	85.000	PERCENT
Base material thickness	7.0002	MM
Corrosion allowance	0.000000	MM
THICKNESSES REQUIRED		
Circumferential stress thickness	1.7198	MM
Long tensile stress thickness	5.0619	MM
Long compressive stress thick	6.7886	MM
TRAY STACK DATA		
Tray material	A285C	
Number of trays	15	
Tray spacing	600.000	MM
Tray thickness	5.0000	MM
Cross sectional area	0.785398	M2
VESSEL SKIRT DATA		
Skirt material	CS	
Skirt height	1.5000	M
Skirt thickness	8.0000	MM
NOZZLE AND MANHOLE DATA		
Nozzle ASA rating	150	CLASS
Nozzle material	SS316	
Nozzle A Quantity	4	
Nozzle A Diameter	80.000	MM DIAM
Nozzle A Location	S	
Nozzle B Quantity	2	
Nozzle B Diameter	200.000	MM DIAM
Nozzle B Location	S	
Nozzle C Quantity	1	
Nozzle C Diameter	250.000	MM DIAM
Nozzle C Location	S	
Nozzle D Quantity	6	
Nozzle D Diameter	50.000	MM DIAM
Nozzle D Location	S	
Number of manholes	2	
Manhole diameter	600.000	MM
PROCESS DESIGN DATA		
Molecular weight Overhead prod	50.000	
WEIGHT DATA		
Shell	3500	KG
Trays and supports	1500	KG
Heads	120	KG
Nozzles	130	KG
Manholes and Large nozzles	580	KG
Skirt	310	KG
Base ring and lugs	120	KG
Ladder clips	90	KG
Platform clips	150	KG
Fittings and miscellaneous	70	KG
Total weight	6600	KG
VENDOR COST DATA		
Cost per unit Trays and supports	1178	DOLLARS
Labor per unit Trays and support	9	MH/TRAY
Material cost	69812	DOLLARS
Shop labor cost	20862	DOLLARS
Shop overhead cost	21618	DOLLARS
Office overhead cost	19090	DOLLARS
Profit	19718	DOLLARS
Total cost	151100	DOLLARS
Cost per unit weight	22.894	USD/KG
Cost per tray	10073	DOLLARS
Cost per unit height or length	7555	USD/M

Table 3: ASPEN Capital Cost Estimator output for recovery column condenser

Summary Costs

Item	Material(USD)	Manpower(USD)	Manhours
Equipment&Setting	151100.	1670.	54
Piping	49127.	21499.	718
Civil	2485.	2983.	121
Structural Steel	12742.	2401.	85
Instrumentation	33440.	13031.	424
Electrical	2746.	1509.	51
Insulation	14133.	12218.	535
Paint	349.	622.	28
Subtotal	266122	55933	2016

Total material and manpower cost=USD 322100.

Total material and manpower cost=USD 50200.

RECOVERY COLUMN CONDENSER

Item Code: EHE JACKETED

Internal Name : EHE JACKETED RECOVERY COLUMN CONDENSER

[Sizing Data](#)

[Design Data](#)

[Summary Costs](#)

Sizing Data

Description	Value	Units
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Design Data

Parameter	Value	Units
Item type	JACKETED	
Material	CS	
Heat transfer area	1.5000	M2
Tube length	6.0003	M
Number of tubes per shell	1	
Design gauge pressure	200.000	KPAG
Temperature	120.000	DEG C
Source of quote	SG	
Number of sections	1	
Total weight	360	KG

Item	Material(USD)	Manpower(USD)	Manhours
Equipment&Setting	2700.	472.	16
Piping	15314.	11040.	367
Civil	942.	1426.	59
Structural Steel	0.	0.	0
Instrumentation	8007.	2152.	69
Electrical	0.	0.	0
Insulation	3571.	3214.	141
Paint	383.	993.	44
Subtotal	30917	19297	696

Total material and manpower cost=USD 50200.

Table 4: DCFROR for 5 year recovery term (Goal Seek Method)

year	Pre-tax CF	depreciation	taxable income	tax due	Cash flow	Fd	Present Value	NPV
1	-12.735	0	-12.735	0	-12.735	0.86713	-11.0429	-11.042895
2	-12.735	0	-12.735	0	-12.735	0.751914	-9.57562	-20.618515
3	3.62	4.226	-0.606	0	3.62	0.652007	2.360264	-18.258251
4	6.36	6.7616	-0.4016	-0.2121	6.5721	0.565374	3.715695	-14.542556
5	6.36	4.05696	2.30304	-0.14056	6.50056	0.490253	3.186916	-11.35564
6	6.36	2.434176	3.925824	0.806064	5.553936	0.425113	2.361048	-8.9945918
7	6.36	2.434176	3.925824	1.374038	4.985962	0.368628	1.837963	-7.1566287
8	6.36	1.217088	5.142912	1.374038	4.985962	0.319648	1.593752	-5.5628765
9	6.36	0	6.36	1.800019	4.559981	0.277176	1.263918	-4.2989587
10	6.36	0	6.36	2.226	4.134	0.240348	0.993597	-3.3053616
11	6.36	0	6.36	2.226	4.134	0.208413	0.861577	-2.4437843
12	6.36	0	6.36	2.226	4.134	0.180721	0.747099	-1.6966851
13	6.36	0	6.36	2.226	4.134	0.156708	0.647832	-1.0488533
14	6.36	0	6.36	2.226	4.134	0.135886	0.561754	-0.4870993
15	6.36	0	6.36	2.226	4.134	0.117831	0.487114	1.426E-05

Table 5: DCFROR for 7-year recovery term (Goal Seek Method)

year	Pre-tax CF	depreciation	taxable income	tax due	Cash flow	Fd	Present Value	NPV
1	-12.735	0	-12.735	0	-12.735	0.8695	-11.07312	-11.073
2	-12.735	0	-12.735	0	-12.735	0.756	-9.628102	-20.701
3	3.62	3.019477	0.600523	0	3.62	0.6574	2.3796943	-18.322
4	6.36	5.174737	1.185263	0.21018305	6.14981695	0.5716	3.5151645	-14.806
5	6.36	3.695637	2.664363	0.41484205	5.94515795	0.497	2.9547295	-11.852
6	6.36	2.639137	3.720863	0.93252705	5.42747295	0.4321	2.3454321	-9.5062
7	6.36	1.886909	4.473091	1.30230205	5.05769795	0.3757	1.9004172	-7.6058
8	6.36	1.886909	4.473091	1.56558185	4.79441815	0.3267	1.5664006	-6.0394
9	6.36	1.886909	4.473091	1.56558185	4.79441815	0.2841	1.3619893	-4.6774
10	6.36	0.942398	5.417602	1.56558185	4.79441815	0.247	1.1842532	-3.4931
11	6.36	0	6.36	1.8961607	4.4638393	0.2148	0.9587118	-2.5344
12	6.36	0	6.36	2.226	4.134	0.1867	0.7720063	-1.7624
13	6.36	0	6.36	2.226	4.134	0.1624	0.6712615	-1.0912
14	6.36	0	6.36	2.226	4.134	0.1412	0.5836636	-0.5075
15	6.36	0	6.36	2.226	4.134	0.1228	0.507497	3.6E-06

Table 6: Sensitivity Analysis

Percent	NPV 5-YEAR (\$MM)			Percent	NPV 7-YEAR (\$MM)		
Increase/Decrease	Product Cost	Raw Material	Utility	Increase/Decrease	Product Cost	Raw Material	Utility
60	0.084810875	3.105932208	1.74570139	60	0.065667306	2.404859064	1.351660477
70	0.16962175	2.525148136	1.639156234	70	0.131334613	1.955169971	1.269164767
80	0.3392435	2.052965965	1.539113835	80	0.262669225	1.589569082	1.191704007
90	0.678487	1.66907802	1.44517731	90	0.52533845	1.292332587	1.118970899
100 (BASE)	1.356974	1.356974	1.356974	100	1.0506769	1.0506769	1.0506769
110	2.035461	1.04486998	1.26877069	110	1.57601535	0.809021213	0.982382902
120	3.0531915	0.804549885	1.186300595	120	2.364023025	0.622946334	0.918528013
130	4.57978725	0.619503411	1.109191056	130	3.546034538	0.479668677	0.858823692
140	6.869680875	0.477017627	1.037093638	140	5.319051806	0.369344881	0.803000152